

Homework #3. Due on Thursday, Sep 17th, 11:59pm on Canvas

Reading for this homework: Sections 1.6 and 2.1.

Plan for next week (Sep 14,16): continue with 2.1 (kernels and images of linear maps, rank-nullity theorem), 2.2 (matrix of a linear transformation), 2.3 (composition of linear maps and matrix multiplication) and start 2.4 (invertibility and isomorphisms).

Problems:

Problem 1: Let V be a vector space over $\mathbb{C}$ with $\dim(V) = n$. As discussed in HW#1, we can also consider V as a vector space over $\mathbb{R}$ (by restricting the scalars from $\mathbb{C}$ to $\mathbb{R}$). Prove that as an $\mathbb{R}$ -vector space V has dimension $2n$. **Hint:** If $\{v_1, \dots, v_n\}$ is a basis of V as a vector space over $\mathbb{C}$, how to get a basis of V as a vector space over $\mathbb{R}$? Use the basic example $V = \mathbb{C}$ to make a guess.

The next several problems in this homework deal with the notion of an internal direct sum of subspaces and the notion of a (vector space) complement. Let V be a vector space and U, W subspaces of V . We say that V is an (internal) direct sum of U and W if

- (1) $V = U + W$ (that is, every element of V can be written as $u + w$ for some $u \in U$ and $w \in W$) and
- (2) $U \cap W = \{0\}$.

If V is a direct sum of U and W , we write $V = U \oplus W$.

Given a subspace U of V , a (vector space) complement of U in V is any subspace W of V such that $V = U \oplus W$ (unlike the set-theoretic complement, such W is almost never unique).

Problem 2: Let V be a vector space, U and W subspaces of V , β_U a basis for U and β_W a basis for W .

- (a) Prove that $V = U + W \iff \beta_U \cup \beta_W$ spans V ;
- (b) Prove that $U \cap W = \{0\}$ if and only if $\beta_U \cap \beta_W = \emptyset$ and $\beta_U \cup \beta_W$ is linearly independent.

Thus, $V = U \oplus W$ if and only if $\beta_U \cap \beta_W = \emptyset$ and $\beta_U \cup \beta_W$ is a basis of V .

Problem 3: Let V be a finite-dimensional vector space and U a subspace of V . Use Problem 2 and a suitable theorem from class to prove that

- (i) U has a complement;
- (ii) for any complement W of U we have $\dim(U) + \dim(W) = \dim(V)$.

Problem 4: Let V be a finite-dimensional vector space and U and W subspaces of V .

- (a) Prove that $\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$. See Problem 29(a) in § 1.6 for a hint (this is a major generalization of 3(ii)).
- (b) Prove that $V = U \oplus W$ if and only if $V = U + W$ and $\dim(U) + \dim(W) = \dim(V)$.
- (c) Prove that $V = U \oplus W$ if and only if $U \cap W = \{0\}$ and $\dim(U) + \dim(W) = \dim(V)$.

Problem 5: Let $V = \mathbb{R}^2$.

- (a) Let $u = (1, 2)$ and $U = \text{Span}(u)$. Find infinitely many different complements for U .
- (b) Let U and W be subspaces of V with $\dim(U) = \dim(W) = 1$. Prove that W is a complement of $U \iff W \neq U$.

Problem 6: Let F be a field, $V = F^n$ and $W = \{(x_1, \dots, x_n) \in V : \sum_{i=1}^n x_i = 0\}$. Let $\{e_1, \dots, e_n\}$ be the standard basis of V (that is, e_i is the vector whose i^{th} coordinate is 1 and all other coordinates are 0).

- (1) Let $U = \text{Span}(e_1)$. Prove (directly from definition) that U is a complement of W .
- (2) Let $S = \{e_i - e_1 : 2 \leq i \leq n\}$. Prove directly from definition that S is both linearly independent and spanning for W (and hence a basis for W).
- (3) Now use (1), an earlier problem in this HW and a theorem from class to explain why to show that S from (b) is a basis for W , it suffices to check that it is either linearly independent or spanning for W (and there is no need to check both).

Problem 7: For each of the following maps T do the following: Prove that T is linear, find a basis for $\text{Ker}(T)$ and $\text{Im}(T)$ and then verify the assertion of the rank-nullity theorem. As usual, $\mathcal{P}_n(F)$ denotes the space of polynomials of degree $\leq n$ over a field F .

- (a) $T : \mathcal{P}_6(\mathbb{R}) \rightarrow \mathcal{P}_6(\mathbb{R})$ given by $T(f(x)) = f'(x)$
- (b) $T : \mathcal{P}_6(\mathbb{Z}_3) \rightarrow \mathcal{P}_6(\mathbb{Z}_3)$ given by $T(f(x)) = f'(x)$ (where as before $\mathbb{Z}_3$ is the field of congruence classes mod 3). **WARNING:** The answer in (b) is different from the answer in (a).
- (c) $T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathbb{R}$ given by $T(f(x)) = f(2)$, that is, T is the evaluation map at $x = 2$.

- (d) $T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_4(\mathbb{R})$ given by $T(f(x)) = (x+1)p(x)$, that is, T is the multiplication by $x+1$.

Problem 8: Let $f : V \rightarrow W$ be a linear map, let $S = \{s_1, \dots, s_n\}$ a finite subset of V and $f(S) = \{f(s_1), \dots, f(s_n)\}$. Determine if the following statements are true (always) or false (in at least one case). If the statement is true, prove it; if false, give a specific counterexample.

- (a) If S spans V , then $f(S)$ spans $\text{Im}(f)$.
- (b) If S is linearly independent and the vectors $f(s_1), \dots, f(s_n)$ are distinct, then $f(S)$ is linearly independent.
- (c) If $f(S)$ spans $\text{Im}(f)$, then S spans V .
- (d) If the vectors $f(s_1), \dots, f(s_n)$ are distinct and $f(S)$ is linearly independent, then S is linearly independent.